3 There are three buses, **A**, **B** and **C**.

The total number of seats in the three buses is 250

28% of the 250 seats are in bus **A**.

number of seats in bus **B** : number of seats in bus **C** = 3 : 2

38 of the seats in bus **B** are empty.
26 of the seats in bus **C** are empty.

Jamie says,

> "The percentage of the seats in bus **B** that are empty is greater than the percentage of the seats in bus **C** that are empty."

Is Jamie correct?
You must show how you get your answer.

Bus A: 28% of 250 = 70 seats

1 mark
Find seats in A.

B and C together: 250 − 70 = 180

1 mark
Subtract from 250.

3 + 2 = 5 parts; one part = 180 ÷ 5 = 36

Bus B: 3 × 36 = 108; bus C: 2 × 36 = 72

1 mark
Share the remaining seats in the ratio 3 : 2.

$\frac{38}{108} \times 100 = 35.185...\%$

$\frac{26}{72} \times 100 = 36.111...\%$

1 mark
Compare fractions of each bus total.

No: 35.185...% < 36.111...%.

1 mark
Bus C has the greater percentage of empty seats.

(Total for Question 3 is 5 marks)

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